

MUAN INTL, Korea, Republic ofWMO: **471630**Lat: **34.983N**Lon: **126.383E**Elev: **52**StdP: **14.67**Time Zone: **9.00 (E09)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	23.2	10.7	9.5	28.0	14.0	11.2	27.2	24.5	30.2	23.0	30.6	8.4	50	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	91.0	79.6	88.1	78.9	86.1	77.8	81.3	86.0	80.4	85.3	79.3	83.9	8.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	158.9	83.4	79.0	151.0	82.8	78.4	148.0	82.5	45.0	87.0	44.2	84.7	43.0	83.6	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	20.0	17.7	16.6	94.5	4.1	1.6	13.6	95.7	11.2	96.6	8.9	97.5	5.9	98.7		
(5)			15.3	82.8	3.9	2.2	12.4	84.4	10.1	85.7	7.9	87.0	5.0	88.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	33.9	37.0	44.3	54.4	64.4	72.1	79.2	80.4	72.4	61.6	50.2	38.4	(7)	(8)
	DBStd	16.81	5.59	6.05	5.91	5.46	4.32	2.98	3.78	4.00	4.10	5.05	6.83	6.24		
	HDD50	1528	500	366	197	21	0	0	0	0	0	1	81	362		
	HDD65	4169	965	783	641	318	65	0	0	0	2	126	445	824		
	CDD50	4261	0	3	21	154	447	664	904	943	672	362	88	3		
	CDD65	1425	0	0	0	1	46	214	439	478	224	22	1	0		
	CDH74	10731	0	0	0	6	229	842	3775	4663	1147	67	2	0		
	CDH80	2996	0	0	0	0	16	82	1092	1619	184	3	0	0		
	WSAvg	7.9	8.4	8.9	9.0	8.3	7.4	6.7	7.6	7.2	6.7	7.5	8.1	8.8		
	PrecAvg	48.3	1.2	1.7	2.7	3.4	3.9	6.1	9.3	9.0	5.3	2.1	2.2	1.5		
(15) Precipitation	PrecMax	69.1	2.6	4.2	5.7	8.3	9.7	15.0	19.5	18.2	17.7	7.8	6.1	3.2	(16)	(17)
	PrecMin	26.9	0.3	0.4	0.9	0.4	1.2	0.8	1.0	0.9	0.7	0.2	0.2	0.2		
	PrecStd	10.2	0.6	0.9	1.4	2.0	1.9	3.3	3.9	4.6	4.3	1.8	1.5	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	62.6	68.0	74.9	82.2	84.5	91.6	93.6	86.4	79.0	71.4	60.6	(20)	(21)
		MCWB	47.8	56.8	58.6	59.9	65.1	70.4	79.8	79.3	74.1	68.8	62.6	55.1		
	2%	DB	48.6	55.8	62.6	71.5	78.8	82.4	89.7	91.5	84.2	75.4	66.5	55.1		
		MCWB	43.6	50.2	55.3	58.2	64.7	70.5	79.2	79.7	73.9	65.8	59.8	49.7		
	5%	DB	46.1	51.5	59.0	68.1	76.8	80.5	87.8	89.7	82.3	73.3	64.1	51.5		
		MCWB	41.9	45.7	52.6	56.7	64.2	69.9	79.0	79.1	72.2	64.0	58.9	47.1		
	10%	DB	42.9	46.7	55.5	64.7	73.8	78.7	85.9	87.8	80.2	70.2	61.0	48.4		
		MCWB	38.7	42.6	50.1	55.5	63.5	69.6	78.2	79.1	71.8	62.8	56.3	43.9		
	0.4%	WB	49.4	58.1	60.3	64.6	72.1	76.0	82.8	82.8	78.5	73.4	64.7	56.6		
		MCDB	51.8	60.8	63.6	70.4	75.0	79.5	86.1	88.0	82.6	77.3	69.6	58.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.2	51.8	56.9	61.6	68.3	74.5	81.3	81.6	76.5	70.0	62.1	50.9	(28)	(29)
		MCDB	47.9	54.3	61.1	67.1	74.0	77.9	85.3	86.8	80.6	72.5	65.0	54.0		
	5%	WB	41.7	46.4	54.0	59.6	66.9	73.2	80.4	80.7	75.1	67.2	59.7	47.4		
		MCDB	44.6	49.8	58.3	64.7	72.5	76.7	84.7	86.0	78.9	69.9	62.8	50.7		
	10%	WB	39.1	42.9	50.7	57.5	65.3	71.9	79.2	79.8	73.5	64.5	57.1	44.5		
		MCDB	42.4	46.0	54.7	62.6	71.3	75.8	83.4	85.0	77.9	69.2	60.9	47.7		
	5% DB	MDBR	13.2	13.8	16.0	17.0	16.9	12.7	10.4	11.6	13.7	15.8	14.8	13.4		
		MCDBR	17.6	20.1	21.6	23.0	20.8	15.6	13.0	14.6	15.9	18.2	16.8	17.3		
		MCWBR	13.1	15.3	14.9	12.1	9.6	6.8	5.7	5.4	7.2	9.9	11.4	13.4		
		MCWBR	16.1	17.8	19.0	16.8	16.1	10.8	11.1	12.6	12.1	12.3	14.1	15.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	13.2	14.8	14.5	11.0	9.4	6.1	5.9	5.5	6.9	9.2	11.3	13.5		
	Clear-Sky Solar Irradiance	taub	0.457	0.497	0.563	0.577	0.595	0.657	0.615	0.562	0.505	0.480	0.468	0.456		
		taud	1.932	1.866	1.736	1.754	1.769	1.717	1.864	2.002	2.089	2.016	1.980	1.943		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	222	229	226	232	230	215	224	233	240	232	217	212	(44)	(45)
		Edh at Noon	47	56	69	71	71	75	64	55	48	47	44	44		
(46) Station Only	RadStd		738	1000	1381	1606	1746	1562	1469	1519	1347	1180	816	652		
			61	93	122	127	191	144	198	209	129	102	87	47		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	N/A

Nomenclature: See separate page